RELATIONALAI INTRO

Bring decision intelligence to high-stakes business questions.

Rel is a decision agent grounded in your semantic model, running inside Snowflake, and powered by RelationalAI reasoners.

PYTHON-BACKED

The model is code you can read, review, test, and deploy.

Concepts, relationships, rules, and curated queries are authored in Python, so the semantic layer behaves like durable software.

Source controlled | Reviewable | Testable

```python
CellTower = model.Concept(
  "CellTower",
  identify_by={"id": String},
)
NetworkEquipment.tower = model.Property(
  f"{NetworkEquipment} installed on "
  f"{CellTower:tower}"
)

model.where(
  CellTower.status == "DEGRADED",
  CellTower.avg_health_score < 0.85,
).define(CellTower.operational_critical_restore())
```

EXTENSIBLE

Add logic without rebuilding the whole system.

Extend the domain

Add concepts like towers, subscribers, advisories, assets, accounts, suppliers, claims, or transactions.

Extend the rules

Encode durable business logic once so future answers use the corrected definition.

Extend the tools

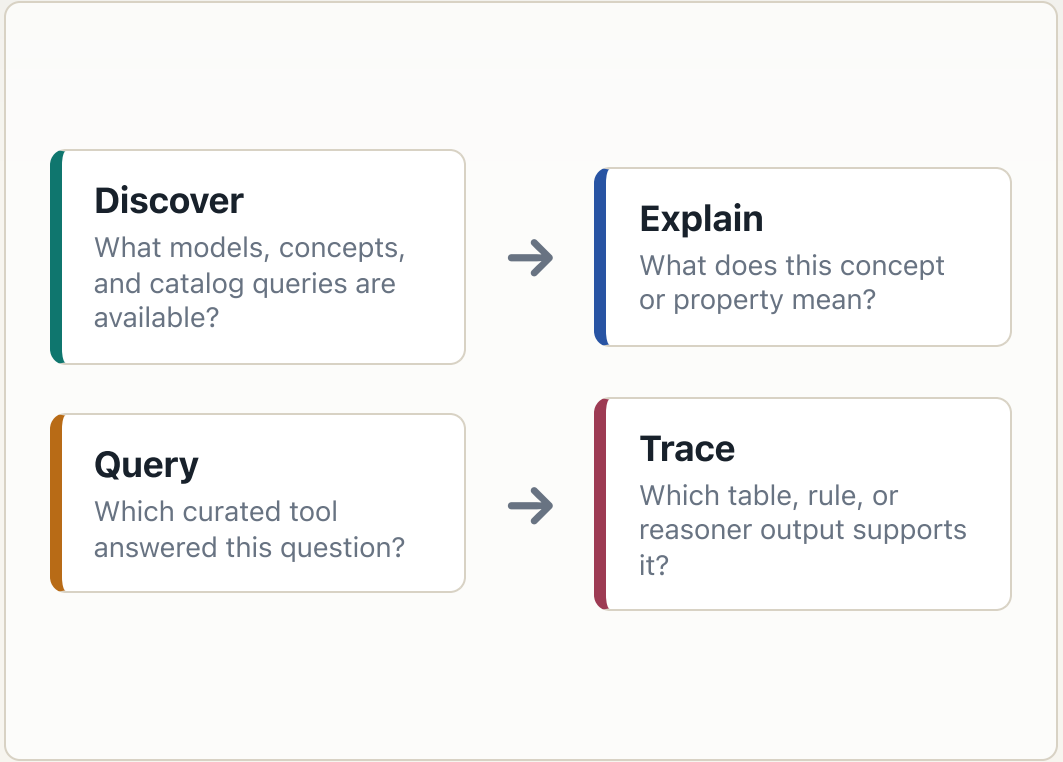
Create curated catalog queries for agents when important questions need reliable answer paths.

Extend the reasoners

Write predictive, graph, path, and optimization outputs back into the model or result tables.

You can inspect the model and every answer surface.

RelationalAI exposes what concepts exist, how tables and documents connect, what queries the agent can run, and what result tables power the answer.



Inspection is part of the workflow

PROVENANCE

Answers are grounded in model surfaces and result artifacts.

Plan summary

Backed by **RESTORE_PLAN** from the optimization stage.

Failure risk

Backed by **TOWER_FAILURE_SCORE** and equipment predictions.

Customer impact

Backed by graph-derived **TOWER_CUSTOMER_IMPACT**.

Call path

Backed by path output in **TOP_CALL_PATH**.

Selected towers

Backed by prescriptive **SELECTED_UPGRADES**.

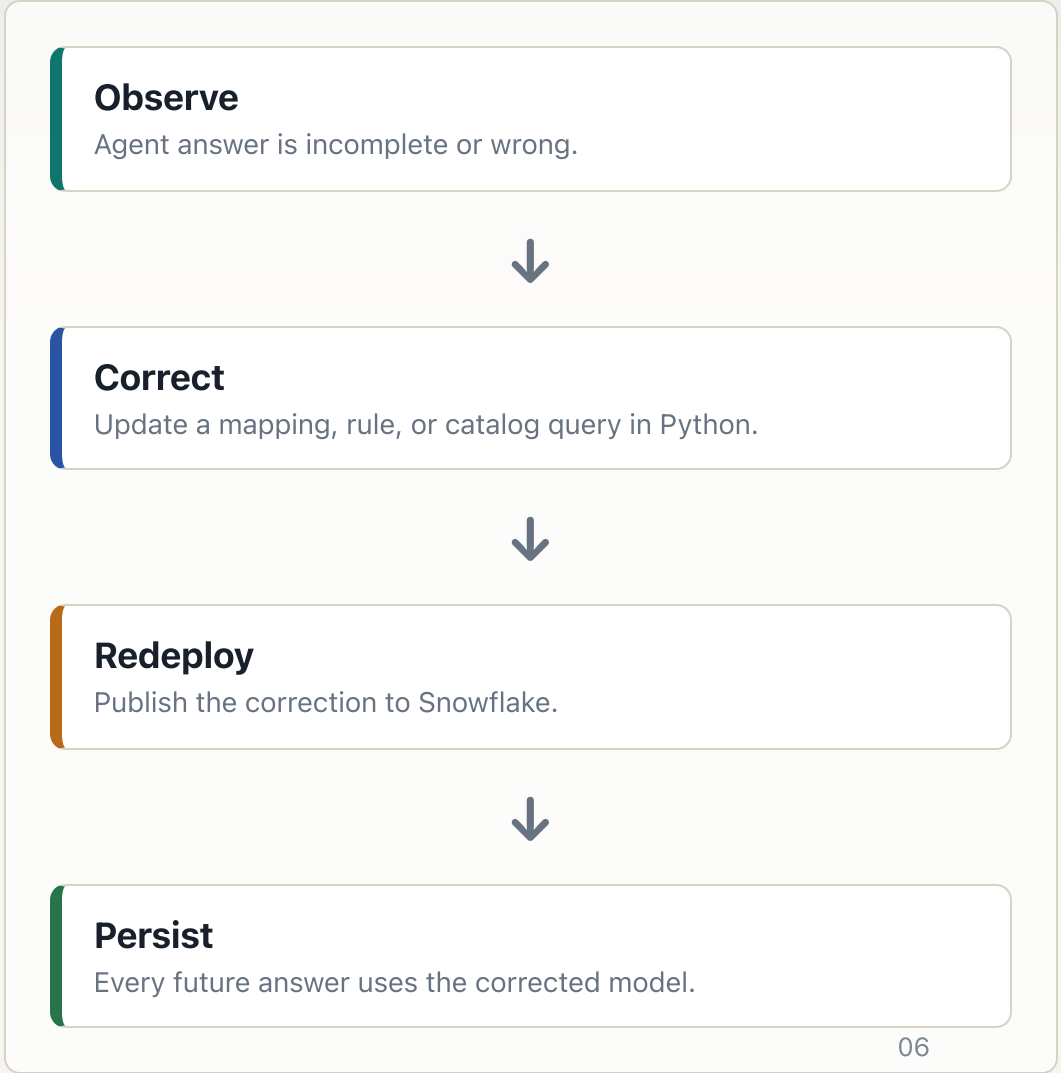
Agent answer

Backed by a catalog query, not an improvised SQL guess.

CORRECTION LOOP

When an answer is wrong, fix the model so the correction persists.

The right unit of correction is not a one-off chat response. It is business intuition turned into a durable rule, relationship, query, or deployed result definition.



Durable corrections beat prompt patches

REASONER WORKFLOW

Multiple reasoners operate over the same business model.

01

Infer with rules

Flag operationally critical towers from status, health, and packet loss.

02

Anticipate risk

Score equipment failure risk from health, tower context, and advisories.

03

Discover impact

Measure customer impact and influence across the subscriber call network.

04

Find paths

Trace influential call routes and the towers that carry each hop.

05

Optimize action

Select upgrade tiers under budget and install-week constraints.

One business question spans the full platform.

It starts as a decision problem, not an algorithm choice.

> Given tower health, equipment advisories, subscriber impact, call graph, and fixed capex, which towers should we upgrade, why those towers, and what constraint limits the plan?

Business users ask in Snowflake; RelationalAI supplies grounded tools.

- What is the final telco recovery plan summary?
- Which towers have the highest combined failure risk and customer impact?
- Show selected upgrades by region.
- Which equipment items are most likely to fail?
- What is the most influential call path and which towers does it use?

Example answer surface

$4.99M
optimized spend

183
Gbps restored

27
towers selected

Budget
binding constraint

The answer is backed by deployed result tables and curated RelationalAI query tools.

AGENT ANSWER 01

Question: What is the final telco recovery plan summary?

The optimized plan restores 183 Gbps across 27 towers with budget as the binding constraint.

Final plan

OPTIMAL

The plan spends just under the $5M capex envelope and uses 167 install weeks.

$4.99M total cost	**183** Gbps restored
27 towers covered	**Budget** binding constraint

TIER	SELECTED TOWERS	INTERPRETATION
Gold	8	High-capacity upgrades concentrated on high-value, high-risk towers.
Silver	8	Balanced capacity gain where cost and install weeks matter.
Bronze	11	Cost-efficient coverage for additional critical towers.
Total	**27**	**Budget is binding; install weeks have slack.**

AGENT ANSWER 02

Question: Which towers have the highest combined failure risk and customer impact?

The highest-priority towers combine predicted failure intensity with customer impact.

Risk + impact

Failure intensity comes from the predictive GNN. Weighted impact comes from customer value and graph-derived call activity.

TWR-0021 has the highest failure intensity at 9.86.

TWR-0114 has the largest weighted customer impact at about 3.13M.

TWR-0086 combines high risk, 39 active subscribers, and strong impact.

TOWER	FAILURE INTENSITY	ACTIVE SUBSCRIBERS	WEIGHTED IMPACT
TWR-0021	9.86	32	324,669
TWR-0030	9.08	26	76,180
TWR-0067	8.05	24	1,466,615
TWR-0086	8.02	39	1,120,168
TWR-0114	6.09	34	3,125,836
TWR-0015	5.06	38	1,987,444

AGENT ANSWER 03

Question: Show selected upgrades by region.

The selected upgrades are explainable by region, spend, capacity, and install effort.

Regional allocation

West receives the largest investment and the most restored capacity. East covers the most selected towers.

WEST
largest spend

EAST
most towers

NORTH
lowest cost

183
total Gbps

REGION	TOWERS	COST	CAPACITY	WEEKS
West	6	$1,733,199	54	49
Central	4	$976,697	32	33
East	7	$924,664	40	35
South	6	$828,525	34	27
North	4	$522,704	23	23
Total	**27**	**$4,985,789**	**183**	**167**

AGENT ANSWER 04

Question: Which equipment items are most likely to fail?

The GNN surfaces equipment items with very high predicted failure probability.

Equipment risk

99%+

The top flagged equipment items all receive a predicted failure label of 1.

The top 50 returned items are all above 99% failure probability.

The highest-scored item is EQP-00361 at 99.23%.

RANK	EQUIPMENT	FAILURE PROBABILITY
1	EQP-00361	99.23%
2	EQP-01386	99.22%
3	EQP-00382	99.22%
4	EQP-00969	99.22%
5	EQP-00729	99.22%
6	EQP-00169	99.22%
7	EQP-00244	99.22%
8	EQP-00167	99.21%

AGENT ANSWER 05

Question: What is the most influential call path and which towers does it use?

The graph/path reasoner traces the most influential call route and the towers carrying it.

Top path

0.00904

Influence load for the top call path from the highest-PageRank hub subscriber.

SUB-CON-00900
hub subscriber

3
tower hops

SUB-CON-00900 → SUB-CON-00814 → SUB-ENT-0038 → SUB-CON-00644

TWR-0094 → TWR-0113 → TWR-0055

The path connects four subscribers over three routed tower hops.

These towers are important because disruption would affect a high-influence communication chain.

CLOSE

An answer you can inspect, trace, correct, redeploy, and trust.

RelationalAI turns a modeled business domain into a governed, extensible answer system for Snowflake.